build-list

2026.08.23 - Strains to construct: 25 doubles + 20 triples

45 new strains. All are built from strains that already exist in the run-4 panel – no new single knockouts are needed, and **every one builds in parallel**: nothing waits on anything else.

The design on this sheet is `capped`. Six selection strategies were compared over the same 39 candidate triples; `capped` is the one being built, and every ID below comes from it. It takes the six best-predicted triples that involve a gene the model has no trigenic training data for, then fills the remaining fourteen from further down the ranking, which holds those genes to six of twenty. The comparison is under "Where this list came from"; the full argument is in [[experiments.W019-echo-crispr-array.next-strains-to-construct]].

Two conventions used throughout:

- **`*` on an ID** means the strain involves `YER079W` or `YLR312C-B`, the two genes with zero trigenic training records. Those predictions are extrapolation.
- **`routes`** is how many already-built doubles a triple can be made from. `1` means one parent only, so a failed cross has no backup until the matching new double is done.

Ordering suggestion: **T01–T15 first**, the fifteen at `routes` 1. T16–T20 each have a second route.

What already exists

From the run-4 order sheet, **12 singles, 13 doubles, 0 triples**:

```
experiments/W019-echo-crispr-array/data/run4_doubles_2026-08-06/
  Single-and-Double-KO-Strains-List-Order.csv
```

Singles s1–s12. Eleven are prediction nodes and all eleven are used below. `s11` YLR313C (SPH1) was built but was never a node in the inference panel, so it is in no predicted triple and is not on this plate.

`fitness`, `boot SE`, `plate SD`: **run 4 measured**, 3 plates, relative to on-plate WT (`run4_strain_bootstrap.csv`). All twelve were measured and all twelve now have a published value to compare against. `YLR104W` (LCL2) previously showed `--`; Costanzo does publish it, and the blank came from `build_reference_smf.py` still listing LCL1 (`YPL056C`) in the s7 slot from the pre-run-4 design. Corrected 2026.08.24.

id	ORF	common	fitness	boot SE	plate SD	Costanzo SMF
s1	YJR060W	CBF1	0.5348	0.0506	0.1076	0.5900
s2	YPL081W	RPS9A	0.9007	0.0515	0.1105	0.9550
s3	YDR057W	YOS9	1.2430	0.1282	0.2690	1.0435
s4	YPL046C	ELC1	1.2066	0.0495	0.1052	1.0433
s5	YBR203W	COS111	1.1292	0.0679	0.1426	1.0370
s6	YGL087C	MMS2	1.1628	0.0329	0.0706	0.9960
s7	YLR104W	LCL2	1.1748	0.0564	0.1187	1.0322
s8	YLL012W	YEH1	1.2604	0.0659	0.1402	0.9925
s9	YER079W	–	1.3333	0.1504	0.3162	1.0387
s10	YKL033W-A	–	0.9251	0.0445	0.0954	1.0327

id	ORF	common	fitness	boot SE	plate SD	Costanzo SMF
s11	YLR313C	SPH1	1.0882	0.1471	0.3083	0.9843
s12	YLR312C-B	–	0.8390	0.0661	0.1397	1.0845

Doubles d1–d13. Eight are parents for the triples below and must be re-measured on this plate; five are not used by any selected triple. `tier` is how the pair was classed when the round was designed: `validation` reproduces a published interaction, `coverage` buys triples, `novel` had never been measured by anyone.

Fitness only, no epsilon. Run 4's epsilon is **not reportable**: the round's denominator rests on a single WT colony and the measured double/single ratio is 0.758 where a multiplicative model wants ~1.07, which no choice of reference can produce. The numbers below are fitness; the diagnosis is in [[experiments.W019-echo-crispr-array.run4-handoff]].

id	pair	used	serves (rank)	fitness	boot SE	plate SD	Costanzo DMF	tier
d1	YDR057W + YPL081W	yes	12, 13, 16	0.6071	0.0479	0.1029	1.0351	coverage
d2	YPL046C + YPL081W	no	–	0.9335	0.0040	0.0084	–	novel
d3	YER079W + YPL081W	no	–	0.9496	0.0556	0.1195	0.8625	validation
d4	YLR312C-B + YPL081W	yes	1, 3, 4, 6	0.6480	0.0454	0.0968	1.0165	coverage
d5	YDR057W + YGL087C	yes	16, 39	0.7884	0.0007	0.0015	1.1372	validation
d6	YDR057W + YER079W	no	–	0.9403	0.0472	0.0991	1.0963	coverage
d7	YDR057W + YLR312C-B	no	–	0.8895	0.0543	0.1150	1.1783	validation
d8	YJR060W + YPL046C	yes	21, 31, 33	0.5649	0.0435	0.0930	0.9663	coverage
d9	YBR203W + YPL046C	yes	24, 28, 29, 31, 34	0.8882	0.0232	0.0501	1.1179	coverage
d10	YDR057W + YLL012W	yes	25, 38, 39	0.9315	0.0269	0.0572	0.9995	coverage
d11	YLL012W + YPL046C	yes	21, 24, 27	0.9087	0.0348	0.0744	1.0251	coverage
d12	YER079W + YJR060W	no	–	0.6886	0.0650	0.1365	0.6100	validation
d13	YER079W + YLR312C-B	yes	2, 5	0.7728	0.0514	0.1084	1.0994	coverage

The 14th designed double, `YKL033W-A x YJR060W`, failed to construct and does not exist.

What is missing, and why

Three gaps, and none of them is a measurement that failed.

A fourth was listed here until 2026.08.24: `YLR104W` (LCL2) as "a single with no published SMF". That was wrong. Costanzo publishes it at 1.0322, and the blank came from our own reference panel, which was assembled for the earlier plate design carrying LCL1 (`YPL056C`) in that slot. Run 4 kept LCL2 and the panel was never rebuilt, so the gap was in our artifact, not in the literature. Same root cause as the 10-gene / 31-target basis being wrong: LCL2 is on the plate, and everything frozen before run 4 assumed it was not.

what	which	why
a double with no published DMF	`YPL046C + YPL081W` (d2)	Tier `novel`: the only one of the 45 candidate pairs with no measurement in Costanzo 2016, Kuzmin 2018 or Kuzmin 2020. There is nothing to compare against because nobody has made it before. It was measured normally, at 0.9335.
a double that was designed but never built	`YKL033W-A x YJR060W`	Reported failed to construct at the 2026-08-11 handover. **Neither the attempt date nor a cause was recorded**, so no reason can be given here beyond that. Both parent singles exist and were measured. It blocks 0 of the 39 target triples, so it costs no trigenic score.
no triples at all	–	Run 4 carried singles and doubles only, and no triple has been constructed in any round. The 20 in Part 2 are the first.

Hypothesis (untested) on the failed cross: `YJR060W` (CBF1) is by a wide margin the sickest single in the panel, 0.5348 against 0.8390 for the next lowest, and it was already the weakest before this round. A cross into a background that slow is a plausible reason for a failure, but nothing was recorded and nothing has been tested. Do not re-propose the pair until an attempt is run and its outcome written down.

Tables and gap rows are generated, not transcribed:

```
experiments/W019-echo-crispr-array/scripts/run4_measured_summary.py
  -> results/run4_measured_summary_singles.csv
  -> results/run4_measured_summary_doubles.csv
  -> results/run4_measured_summary_gaps.csv
```

Part 1 – 25 double knockouts

Cross the two singles listed. Both parents already exist.

ID	gene 1	gene 2	ID	gene 1	gene 2
D01	YBR203W (COS111)	YDR057W (YOS9)	D14	YGL087C (MMS2)	YPL046C (ELC1)
D02	YBR203W (COS111)	YGL087C (MMS2)	D15	YGL087C (MMS2)	YPL081W (RPS9A)
D03	YBR203W (COS111)	YJR060W (CBF1)	D16	YJR060W (CBF1)	YLL012W (YEH1)
D04	YBR203W (COS111)	YKL033W-A	D17	YJR060W (CBF1)	YLR104W (LCL2)
D05	YBR203W (COS111)	YLL012W (YEH1)	D18	YKL033W-A	YLL012W (YEH1)
D06	YBR203W (COS111)	YLR104W (LCL2)	D19*	YKL033W-A	YLR312C-B
D07*	YBR203W (COS111)	YLR312C-B	D20	YKL033W-A	YPL046C (ELC1)
D08	YBR203W (COS111)	YPL081W (RPS9A)	D21	YKL033W-A	YPL081W (RPS9A)

ID	gene 1	gene 2	ID	gene 1	gene 2
D09	YDR057W (YOS9)	YKL033W-A	D22*	YLL012W (YEH1)	YLR312C-B
D10*	YER079W	YLL012W (YEH1)	D23*	YLR104W (LCL2)	YLR312C-B
D11*	YER079W	YLR104W (LCL2)	D24	YLR104W (LCL2)	YPL046C (ELC1)
D12	YGL087C (MMS2)	YLL012W (YEH1)	D25	YLR104W (LCL2)	YPL081W (RPS9A)
D13*	YGL087C (MMS2)	YLR312C-B			

Part 2 – 20 triple knockouts

`build from` is an **existing** double from run 4; cross it with the single in `x third single`. `routes` counts the already-built doubles this triple could be made from; start with the fifteen at `1`.

ID	gene 1	gene 2	gene 3	build from	x third single	routes
T01*	YKL033W-A	YLR312C-B	YPL081W (RPS9A)	YLR312C-B + YPL081W	YKL033W-A	1
T02*	YER079W	YLL012W (YEH1)	YLR312C-B	YER079W + YLR312C-B	YLL012W (YEH1)	1
T03*	YLR104W (LCL2)	YLR312C-B	YPL081W (RPS9A)	YLR312C-B + YPL081W	YLR104W (LCL2)	1
T04*	YBR203W (COS111)	YLR312C-B	YPL081W (RPS9A)	YLR312C-B + YPL081W	YBR203W (COS111)	1
T05*	YER079W	YLR104W (LCL2)	YLR312C-B	YER079W + YLR312C-B	YLR104W (LCL2)	1
T06*	YGL087C (MMS2)	YLR312C-B	YPL081W (RPS9A)	YLR312C-B + YPL081W	YGL087C (MMS2)	1
T07	YBR203W (COS111)	YDR057W (YOS9)	YPL081W (RPS9A)	YDR057W + YPL081W	YBR203W (COS111)	1
T08	YDR057W (YOS9)	YKL033W-A	YPL081W (RPS9A)	YDR057W + YPL081W	YKL033W-A	1
T09	YDR057W (YOS9)	YKL033W-A	YLL012W (YEH1)	YDR057W + YLL012W	YKL033W-A	1
T10	YGL087C (MMS2)	YLL012W (YEH1)	YPL046C (ELC1)	YLL012W + YPL046C	YGL087C (MMS2)	1
T11	YBR203W (COS111)	YLR104W (LCL2)	YPL046C (ELC1)	YBR203W + YPL046C	YLR104W (LCL2)	1
T12	YBR203W (COS111)	YKL033W-A	YPL046C (ELC1)	YBR203W + YPL046C	YKL033W-A	1
T13	YJR060W (CBF1)	YLR104W (LCL2)	YPL046C (ELC1)	YJR060W + YPL046C	YLR104W (LCL2)	1
T14	YBR203W (COS111)	YGL087C (MMS2)	YPL046C (ELC1)	YBR203W + YPL046C	YGL087C (MMS2)	1
T15	YBR203W (COS111)	YDR057W (YOS9)	YLL012W (YEH1)	YDR057W + YLL012W	YBR203W (COS111)	1
T16	YDR057W (YOS9)	YGL087C (MMS2)	YPL081W (RPS9A)	YDR057W + YGL087C	YPL081W (RPS9A)	2
T17	YJR060W (CBF1)	YLL012W (YEH1)	YPL046C (ELC1)	YJR060W + YPL046C	YLL012W (YEH1)	2
T18	YBR203W (COS111)	YLL012W (YEH1)	YPL046C (ELC1)	YBR203W + YPL046C	YLL012W (YEH1)	2
T19	YBR203W (COS111)	YJR060W (CBF1)	YPL046C (ELC1)	YBR203W + YPL046C	YJR060W (CBF1)	2
T20	YDR057W (YOS9)	YGL087C (MMS2)	YLL012W (YEH1)	YDR057W + YGL087C	YLL012W (YEH1)	2

*** on an ID = involves a gene absent from our trigenic library**

`YER079W` and `YLR312C-B` have **zero** trigenic records in the datasets the prediction model was trained on, so predictions involving them are extrapolation. Exposure is deliberately limited to **6 of 20 triples** (T01, T02, T03, T04, T05, T06) and **7 of 25 doubles** (D07, D10, D11, D13, D19, D22, D23).

If those two genes are later dropped, **14 triples survive and 17 of the 25 new doubles are still needed** (D25, `YLR104W + YPL081W`, would then feed no triple). Worth settling before starting the starred strains.

Where this list came from

There were 39 candidate triples. The figure shows which of them each of the six strategies would pick, strongest-predicted at the top. **Read the `capped` column: that is this sheet.** It takes the six best-predicted triples that involve a starred gene, then fills the rest from further down the list to hold the starred genes to six. `rank` would have put 17 of 20 on those genes and `balanced` 15; `capped` is the only design that holds the number to a chosen value while still covering all eleven genes.

How often each gene ends up in a triple, again reading the `capped` bars. It spreads across all eleven genes, range 2 to 8, rather than concentrating on a few.

Notes for the bench

- **Eleven singles are used** and all already exist: YBR203W (COS111), YDR057W (YOS9), YER079W, YGL087C (MMS2), YJR060W (CBF1), YKL033W-A, YLL012W (YEH1), YLR104W (LCL2), YLR312C-B, YPL046C (ELC1), YPL081W (RPS9A).
- **Do not attempt `YKL033W-A x YJR060W`.** It failed to construct previously and is deliberately not in this list. If anyone retries it, please record the attempt date and outcome – we have no record of when the original attempt happened.
- **`YLR104W` (LCL2) and `YKL033W-A` both have no doubles in the current panel.** Six of the new doubles pair with YLR104W (D06, D11, D17, D23, D24, D25) and six with YKL033W-A (D04, D09, D18, D19, D20, D21); those are the first coverage either gene has had. YKL033W-A is at zero only because its one designed double is the pair that failed.
- **`YLR312C-B`** is a merged/deprecated ORF; there is an open question about whether to use it or the adjacent real gene SPH1 (YLR313C). It appears in D07, D13, D19, D22, D23 and among the starred triples.
- The plating round also needs the existing singles plus **8 of the 13 existing doubles** as parents.

Plate

	count
singles	11
doubles	33 (8 existing + 25 new)
triples	20
WT (BY4741)	1
on plate	**65**
to construct	**45**

5 wells per strain on one 384 layout; 65 tubes at one pick per strain.

Why all 65 and not just the 20 triples. A trigenic interaction consumes seven measured fitnesses:

$$\tau_{abc} = f_{abc} - f_{ab}f_c - f_{ac}f_b - f_{bc}f_a + 2f_af_bf_c$$

so every triple needs its own fitness plus **all three** of its doubles and **all three** of its singles on the same plate. Verified against the selection: after the 45 strains are built, all 20 triples have all six supporting terms present, and the 65 strains listed here are exactly the closure of that requirement – no extra, none missing. What is bought is 20 computable tau values, not 20 fitness numbers.

Full rationale and measurement caveats: [[experiments.W019-echo-crispr-array.next-strains-to-construct]]

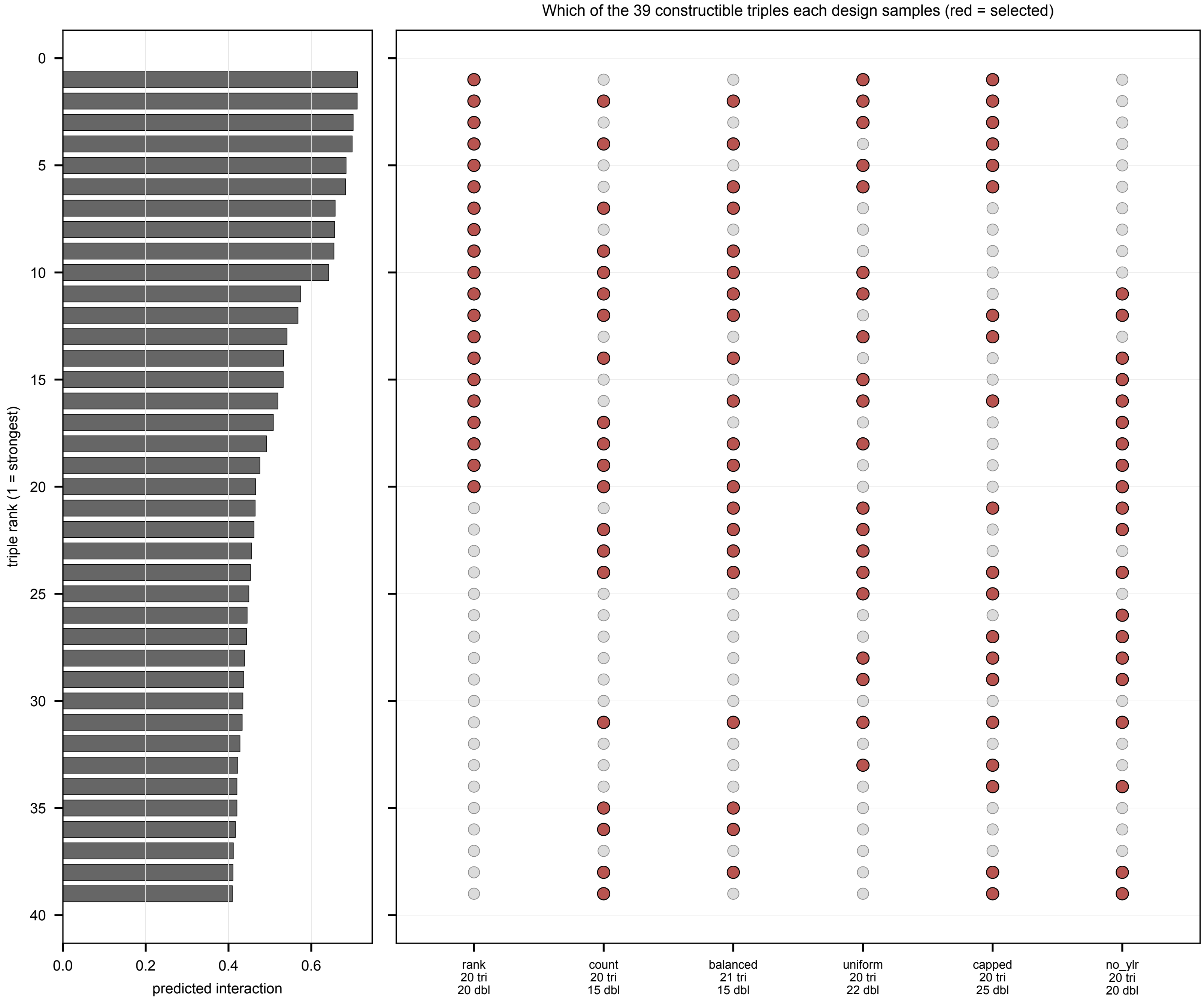

Figure 1: Which of the 39 candidate triples each design selects

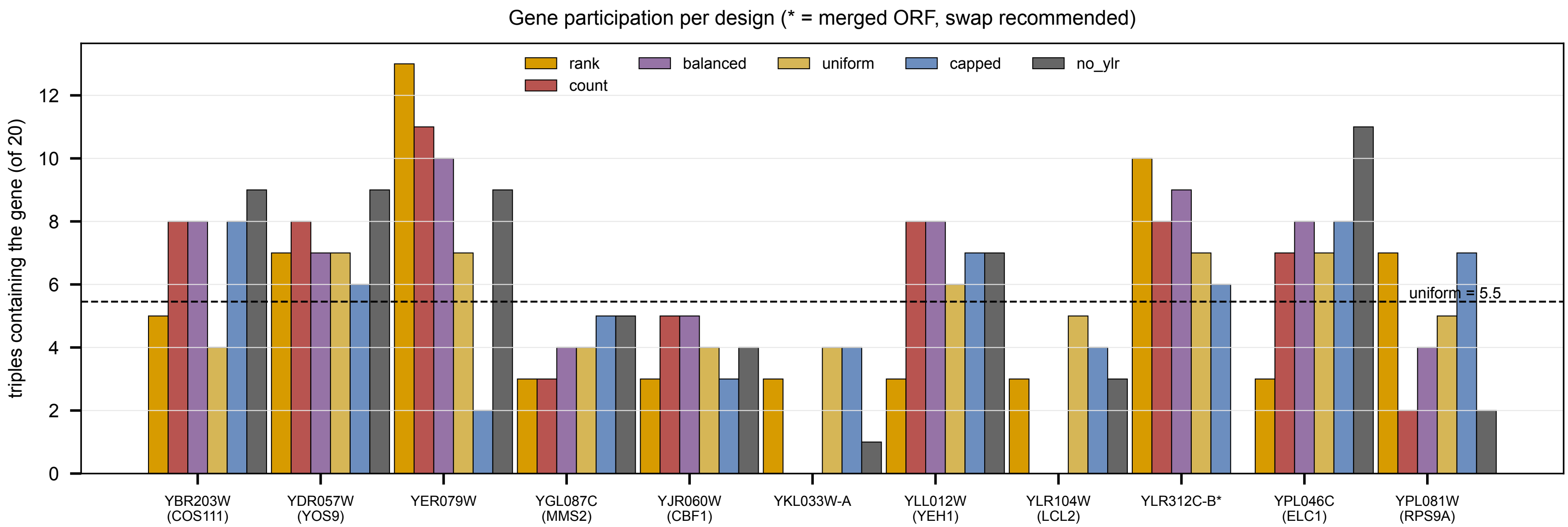

Figure 2: How often each gene appears